量子力学特論 I { Green 関数 / 線形応答理論 / 応用(多体問題)・化学反応(量子カオス)

No. 1

Date 94・10・5

多体系の理論

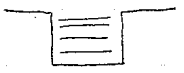
§1 才2量子化

Fermi-Dirac 統計 、Fermi粒子　粒子の入れかえに対して反対称

Bose-Einstein 統計　Bose粒子　〃　対称

多体系の波動関数

$\phi_i(\vec{r})$ $(i=1,2,\cdots)$ single particle wave function a set を選び ある順に並べる



多体系の状態をそれらの状態 (ϕ_i) の粒子の占有数で表す

$\Psi(n_1, n_2, \cdots, n_i \cdots) = |n_1, n_2, \cdots, n_i, \cdots\rangle$

↑ ϕ_i の占有数

$$\langle n_1', n_2', \cdots | n_1, n_2, \cdots \rangle = \delta_{n_1 n_1'} \delta_{n_2 n_2'} \cdots \delta_{n_i n_i'} \cdots$$

For bosens

$$b_i \Psi(\cdots, n_i, \cdots) = \sqrt{n_i}\, \Psi(\cdots, n_i - 1, \cdots)$$

$$b_i^+ \Psi(\cdots, n_i, \cdots) = \sqrt{n_i + 1}\, \Psi(\cdots, n_i + 1, \cdots)$$

$$[b_i, b_j] = [b_i^+, b_j^+] = 0$$

$$[b_i, b_j^+] = \delta_{ij}$$

↳ $[b_i, b_j^+]\Psi = (\sqrt{n_i}\sqrt{n_i} - \sqrt{n_i}\sqrt{n_i})\Psi = \Psi$

For fermion

$$c_i \Psi(\cdots, n_i, \cdots) = \sqrt{n_i}\,(-1)^{\sum_{j<i} n_j}\, \Psi(\cdots, n_i - 1, \cdots)$$

$$c_i^+ \Psi(\cdots, n_i, \cdots) = \sqrt{1 - n_i}\,(-1)^{\sum_{j<i} n_j}\, \Psi(\cdots, n_i + 1, \cdots)$$

$$\{c_i, c_j\} = \{c_i^+, c_j^+\} = 0$$

$$\{c_i, c_j^+\} = \delta_{ij}$$

Slater 行列式

$$\Psi = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(r_1) & \cdots & \phi_i(r_1) & \cdots & \phi_N(r_1) \\ \phi_1(r_2) & \cdots & \phi_i(r_2) & \cdots & \phi_N(r_2) \\ \vdots & & \vdots & & \vdots \\ \phi_1(r_N) & \cdots & \phi_i(r_N) & \cdots & \phi_N(r_N) \end{vmatrix}$$

1体の演算子 single particle operator

$$F_1 = \sum_i f(r_i)$$

$$F_1 = \sum_{i,j} \langle i|f_1|j\rangle\, a_i^+ a_j \qquad \langle i|f|j\rangle = \int \phi_i^*(\vec{r})\, f_1(\vec{r})\, \phi_j(\vec{r})\, d\vec{r}$$

Two-particle operator

$$F_2 = \sum_{i \neq j} f_2(\vec{r}_i, \vec{r}_j)$$

$$F_2 = \frac{1}{2} \sum_{\substack{i,j \\ k,m}} \langle i,j|f_2|k,m\rangle\, a_i^+ a_j^+ a_k a_m \qquad \langle ij|f|km\rangle = \iint d\vec{r}_1 d\vec{r}_2\, \phi_i^*(\vec{r}_1) \phi_j^*(\vec{r}_2)\, f_2(\vec{r}_1, \vec{r}_2)\, \phi_k(\vec{r}_1)\, \phi_m(\vec{r}_2)$$

No. 2

Date 94 · 10 · 5

field operator (bosen)

$\psi_B(\vec{r}) = \sum_i \phi_i(\vec{r})\, b_i$ $\qquad [\psi_B(r), \psi_B(r')] = [\psi_B^\dagger(r), \psi_B^\dagger(r')] = 0$

$\psi_B^\dagger(\vec{r}) = \sum_i \phi_i^*(\vec{r})\, b_i^\dagger$ $\qquad [\psi_B(r), \psi_B^\dagger(r')] = \sum_i \phi_i(r)\phi_i^*(r') = \delta(\vec{r}-\vec{r}')$

(fermion)

$\psi_F(\vec{r}) = \sum_i \phi_i c_i$ $\qquad \{\psi_F(r), \psi_F(r')\} = \{\psi_F^\dagger(r), \psi_F^\dagger(r')\} = 0$

$\psi_F^\dagger(\vec{r}) = \sum_i \phi_i^* c_i^\dagger$ $\qquad \{\psi_F(r), \psi_F^\dagger(r')\} = \sum_{i,j} (\phi_i c_i \phi_j^* c_j^\dagger + \phi_j^* c_j^\dagger \phi_i c_i)$

$= \sum_{i,j} \phi_i \phi_j^* \{c_i, c_j^\dagger\} = \sum_{i,j} \phi_i \phi_j^* \delta_{ij} = \sum_i \phi_i(r)\phi_i(r')$

$= \delta(\vec{r}-\vec{r}')$

number operators

$b_i^\dagger b_i \, \Phi = n_i \Phi$

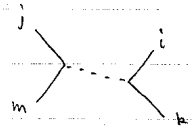
$c_i^\dagger c_i \, \Phi = n_i \Phi$

density operators

$\rho(\vec{r}) = \psi^*(r)\psi(r)$

$\int \phi^*(\vec{r}) f_1(\vec{r}) \phi(\vec{r})\, d\vec{r}$

$F_1 = \int d\vec{r}\, \rho(\vec{r}) f(\vec{r}) = \int \psi^\dagger(\vec{r}) f_1 \psi(\vec{r})\, d\vec{r} = \sum_{i,j} \langle i|f|j\rangle a_i^\dagger a_j$



$F_2 = \sum_{i<j} f(\vec{r}_1, \vec{r}_2) = \frac{1}{2}\int \rho(\vec{r}_1) f(\vec{r}_1, \vec{r}_2) \rho(\vec{r}_2)\, d\vec{r}_1 d\vec{r}_2$

$= \frac{1}{2}\int \psi^\dagger(\vec{r}_1)\psi(\vec{r}_1) f(\vec{r}_1, \vec{r}_2) \psi^\dagger(r_2)\psi(r_2)\, d\vec{r}_1 d\vec{r}_2$

$= \frac{1}{2}\int (\sum_i \phi_i^*(r_i) a_i^\dagger) \cdots$

$= \langle ij|f|km\rangle a_i^\dagger a_j^\dagger a_k a_m$

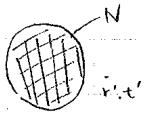
Green function

$T=0$

Heisenberg 表示の $\psi_\sigma^\dagger(\vec{r},t)$, $\psi_\sigma(\vec{r},t)$

$\psi_\sigma^\dagger(r,t) = e^{iHt}\psi^\dagger(r)e^{-iHt}$

$\psi_\sigma(r,t) = e^{iHt}\psi(r)e^{-iHt}$



fermi 球

$x = (\vec{r}, t)$

$G(x,x') = G(\vec{r},t;\vec{r}',t')$

$G_{\sigma\sigma'}(x,x') = -i\langle \Phi_0 | T\{\psi_\sigma(r,t)\psi_{\sigma'}^\dagger(r',t')\} | \Phi_0\rangle$

(Φ_0: N粒子系の基底状態)

$$T\{\psi_\sigma(r,t)\psi_{\sigma'}^\dagger(r',t')\} = \begin{cases} \psi_\sigma(r,t)\psi_{\sigma'}^\dagger(r',t') & t' < t \quad \text{particle の伝播 (propagation)} \\ \mp\psi_{\sigma'}^\dagger(r',t')\psi_\sigma(r,t) & t' > t \end{cases}$$

(− fermion, + bosen)

以下 Fermion を考える

$\langle \Phi_0 | \psi_\sigma(r,t) \underbrace{\psi_{\sigma'}^+(r',t') | \Phi_0\rangle}_{N+1\ 粒子}$ $\quad t' < t$

$\psi_\sigma^+(r,t)|\Phi_0\rangle$ $\quad N-1$ ホール $\quad t' > t$

$N+1$

$\vec{r}$ を Fourier 変換して

$$G(\vec{r},t,\vec{r}',t') = \frac{1}{\Omega}\sum_{k,k'} G(k,t,k',t')\, e^{i(\vec{k}\cdot\vec{r} - \vec{k}'\cdot\vec{r}')}$$

(Ω: volume)

$$G(k,t,k',t') = \frac{1}{\Omega}\iint dr\,dr'\, G(r,t,r',t')\, e^{-i(k\cdot r - k'\cdot r')}$$

$$= -i\langle \Phi_0 | T\{a_k(t)\, a_{k'}^+(t')\} | \Phi_0\rangle$$

$$a_k(t) = \frac{1}{\sqrt{\Omega}}\int dr\, \psi(r,t) e^{-ikr}$$

$$a_k^+(t) = \frac{1}{\sqrt{\Omega}}\int dr\, \psi^+(r,t) e^{ikr}$$

一様な系では

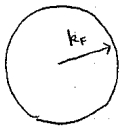
$$\int e^{-ik\cdot r + ik'\cdot r'} dr\,dr' \to \int e^{-i(k-k')\cdot r - ik'(\vec{r}-\vec{r}')} dr \quad (\vec{\rho} = \vec{r} - \vec{r}')$$

$$\to \delta_{k,k'} e^{-i\vec{k}\cdot\vec{\rho}}$$

$$G(k,t) = G(k\,t\,k\,0) = -i\langle \Phi_0 | T\{a_k(t)\, a_k^+(0)\} | \Phi_0\rangle$$

$$= \int dr\, G(\vec{r},t) e^{-i\vec{k}\cdot\vec{r}} \qquad G(\vec{r},t) = \frac{1}{\Omega}\sum_k e^{ik\cdot r} G(\vec{k},t)$$

$\langle \Phi_0 | a_k^+ a_k | \Phi_0\rangle = m_k$ とおく



Fermi 球

$k < k_F \quad m_k = 1$

$k > k_F \quad m_k = 0$

$$G(k,+0) = -i\langle \Phi_0 | a_k(+0)\, a_k^+ | \Phi_0\rangle = -i(1-m_k)$$

($1 - a_k^+ a_k$)

$$G(k,-0) = +i\langle \Phi_0 | a_k^+ a_k | \Phi_0\rangle = i m_k$$

$$\therefore \quad G(k,+0) - G(k,-0) = -i$$

さらに時間 t で Fourier 変換して

$$G(k\omega, k'\omega') = \frac{1}{2\pi}\iint_{-\infty}^{+\infty} dt\,dt'\, G(kt,k't') e^{i(\omega t - \omega' t')}$$

$-i(\omega-\omega')t + i\omega'\tau$

$t' = t + \tau$

$$G(kt,k't') = \frac{1}{2\pi}\iint d\omega\, d\omega'\, G(k\omega,k'\omega') e^{-i(\omega t - \omega' t')}$$

時間的に一様とすると

$$G(k\omega, k'\omega') = G(k\omega)\,\delta_{kk'}\,\delta(\omega-\omega')$$

$$G(k,t) = \frac{1}{2\pi}\int d\omega\, G(k,\omega)\, e^{-i\omega t}$$

$$G(k,\omega) = \int_{-\infty}^{\infty} dt\, G(k,t)\, e^{-i\omega t}$$

例として free electron 系

$$\mathcal{H} = \sum_{k,\sigma} \varepsilon_k^0\, a_{k\sigma}^\dagger a_{k\sigma} \qquad a_k(t) = e^{iHt} a_k e^{-iHt} = e^{-i\varepsilon_k^0 t} a_k$$

$$a_k^\dagger(t) = e^{iHt} a_k^\dagger e^{-iHt} = e^{i\varepsilon_k^0 t} a_k^\dagger$$

$|k| < k_F$

$$G_0(k,t) = \begin{cases} i e^{-i\varepsilon_k^0 t} & t<0 \\ 0 & t>0 \end{cases}$$

$(t<0)$ $\qquad |k| < k_F$

$$G_0(k,t) = i\langle \Phi_0 | a_k^\dagger e^{-i\varepsilon_k^0 t} a_k | \Phi_0\rangle = -i e^{-i\varepsilon_k^0 t}$$

$|k| > k_F$

$$G_0(k,t) = \begin{cases} 0 & t<0 \\ -i e^{-i\varepsilon_k^0 t} & t>0 \end{cases}$$

$(t>0)$

$$-i\langle \Phi_0 | T\, a_k(t) a_k^\dagger | \Phi_0\rangle$$

← Fermi 粒子

$$-i\langle \Phi_0 | a_k(t)\, a_k^\dagger | \Phi_0\rangle$$

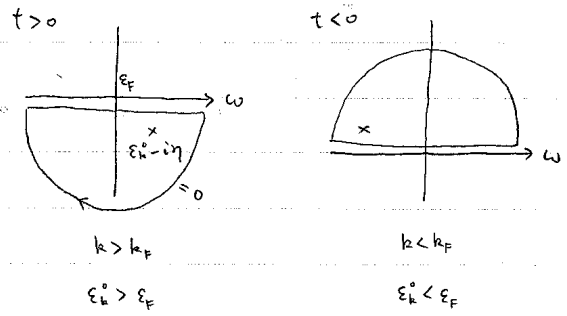
$$e^{-i\varepsilon_k^0 t} a_k a_k^\dagger |\Phi_0\rangle$$

$$G_0(k\omega) = \frac{1}{\omega - \varepsilon_k^0 + i\eta\, \mathrm{sgn}(|k|-k_F)}$$

$$\eta\,\mathrm{sgn}(|k|-k_F) = \begin{cases} +0 & |k|>k_F \\ -0 & |k|<k_F \end{cases}$$

$$G(k,t) = \frac{1}{2\pi}\int d\omega\, G(k\omega) e^{-i\omega t}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega\, \frac{1}{\omega - \varepsilon_k^0 + i\eta\,\mathrm{sgn}(|k|-k_F)}\, e^{-i\omega t}$$



$k > k_F$ $\qquad \varepsilon_k^0 > \varepsilon_F$

$k < k_F$ $\qquad \varepsilon_k^0 < \varepsilon_F$

スペクトル表示

$|\Phi_n\rangle$ E_n $\quad N+1$ 個の粒子系の固有関数とする

$|\Phi_m\rangle$ E_m $\quad N-1$ 〃

$\omega_{n0} = E_n - E_0$

↑ $N+1$ 個系の基底エネルギー

$\omega_{m0} = E_m - E_0$

$$G(k,t) = \begin{cases} -i \sum_n |\langle \Psi_n | a_k^+ | \Psi_0 \rangle|^2 e^{-i\omega_{n0} t} & t > 0 \\ i \sum_m |\langle \Psi_m | a_k | \Psi_0 \rangle|^2 e^{i\omega_{n0} t} & t < 0 \end{cases}$$

$$\langle \Psi_0 | e^{iHt} a_k e^{-iHt} a_k^+ | \Psi_0 \rangle = \langle \Psi_0 | e^{iE_0 t} a_k e^{-iE_n t} | \Psi_n \rangle \langle \Psi_n | a_k^+ | \Psi_0 \rangle$$

$$\uparrow \; |\Psi_n\rangle\langle\Psi_n|$$

$$\langle \Psi_0 | a_k^+ e^{iHt} a_k e^{-iHt} | \Psi_0 \rangle = \langle \Psi_0 | a_k^+ | \Psi_m \rangle \langle \Psi_m | e^{iE_m t} a_k e^{-iE_0 t} | \Psi_0 \rangle$$

$$\uparrow \; |\Psi_m\rangle\langle\Psi_m|$$

μ chemical potential

$E_0(N+1) - E_0(N) = E_0(N) - E_0(N-1) = \mu$

$\omega_{n0} = \mu + \xi_{n0}$

$\omega_{m0} = -\mu + \xi_{m0}$

$A_+(k,\omega) = \sum_n |\langle \Psi_n | a_k^+ | \Psi_0 \rangle|^2 \delta(\omega - \xi_{n0})$

$A_-(k,\omega) = \sum_m |\langle \Psi_m | a_k | \Psi_0 \rangle|^2 \delta(\omega - \xi_{m0})$

$$G(k,\omega) = \int_{-\infty}^{\infty} G(k,t) e^{i\omega t} dt$$

$$G(k,t) = \begin{cases} -i \int_0^\infty A_+(k,\omega') e^{-i(\mu+\omega')t} d\omega' & t > 0 \\ i \int_0^\infty A_-(k,\omega') e^{-i(\mu-\omega')t} d\omega' & t < 0 \end{cases}$$

$$\int_0^\infty -i \int_0^\infty A_+(k,\omega') e^{-i(\mu+\omega'-\omega)t} d\omega' dt$$

$$+ \int_{-\infty}^0 i \int_0^\infty A_-(k,\omega') e^{-i(\mu-\omega'-\omega)t} d\omega' dt$$

$$G(k,\omega) = \int_0^\infty \left\{ \frac{A_+(k,\omega')}{\omega - \omega' - \mu + i\eta} + \frac{A_-(k,\omega')}{\omega + \omega' - \mu - i\eta} \right\} d\omega'$$

No.

Date

$$G(k,\omega)=\int_0^\infty d\omega'\left\{\frac{A_+(k,\omega')}{\omega-\omega'-\mu+i\eta}+\frac{A_-(k,\omega')}{\omega+\omega'-\mu-i\eta}\right\}$$

$\eta\to+0$

$$\begin{cases} A_+(k,\omega)=\sum_n |\langle \Psi_n|a_k^+|\Psi_0\rangle|^2\,\delta(\omega-\xi_{n0}) \\ \qquad E_n^{N+1}-E_0^N=\mu+\xi_{n0} \\ A_-(k,\omega)=\sum_m |\langle \Psi_m|a_k|\Psi_0\rangle|^2\,\delta(\omega-\xi_{m0}) \\ \qquad E_m^{N-1}-E_0^N=-\mu+\xi_{m0} \end{cases}$$

$$\int_{-\infty}^{\infty}A(k,\omega)d\omega=\sum_n|\langle\Psi_n|a_k^+|\Psi_0\rangle|^2+\sum_m|\langle\Psi_m|a_k|\Psi_0\rangle|^2$$

$$=\langle\Psi_0|a_ka_k^++a_k^+a_k|\Psi_0\rangle=1$$

$$A(k,\omega)=A_+(k,\omega)+A_-(k,-\omega)$$

$$A_-(k,-\omega)=\sum_m|\langle\Psi_m|a_k|\Psi_0\rangle|^2\,\delta(\omega+\xi_{m0})$$

$$G(k,\omega)=\int_{-\infty}^{\infty}\frac{A(k,\omega)}{\omega-\omega'-\mu+i\omega'\eta}$$

$$\mathrm{Im}\,G(k,\omega)=\mp\pi A(k,\omega-\mu)\qquad \begin{matrix}-&\omega-\mu>0\\+&\omega-\mu<0\end{matrix}$$

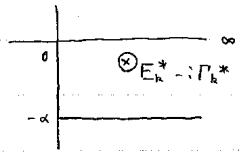
$$\mathrm{Im}\,G(k,\omega+\mu)=\mp\pi A(k,\omega)\qquad \begin{matrix}-&\omega>0\\+&\omega<0\end{matrix}\ =A_-(k,-\omega)$$

$$G(k,\omega)=\int_{-\infty}^{\infty}d\omega'\frac{A(k,\omega)}{\omega-\omega'-\mu+i\omega'\eta}=\int_C\frac{A(k,z)}{\omega-\mu-z}dz\qquad \omega'-i\omega'\eta=z$$



$t>0$ 一粒子をつけ加える $a_k^+|\Psi_0\rangle$

$$\langle\Psi_0|a_k(t)\,a_k^+|\Psi_0\rangle=iG(k,t)=e^{-i\mu t}\underbrace{\int_0^\infty A_+(k,\omega)e^{-i\omega t}d\omega}$$



$$\simeq\int_0^{-i\alpha}A_+(k,\omega)e^{-i\omega t}d\omega$$

$$-2\pi i\,z_k\,e^{-iE_k^*t-\Gamma_k^*t}$$

↑ 留数

$e^{-\alpha t}\to0$ となる α

a) t が小さい

$\alpha t\gg1$ となるためには α は大きい

$A_+(k,\omega)$ の全領域が効く

b) t が大きく $t>\Gamma_k^{*-1}$

$\alpha<\Gamma_k^*$ 準粒子が減衰してる、という

$A(k,\omega)$ の ω の小さい領域が効く

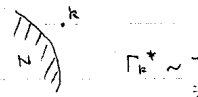
c) $t<\Gamma_k^{*-1}=\tau$

準粒子の寿命より短い

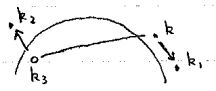
$$iG(k,t)\simeq-2\pi i\,z_k\,e^{(-iE_k^*-i\mu-\Gamma_k^*)t}$$

$$Q_k^+|\Psi_0\rangle=\sqrt{z_k}\left\{a_k^++c_3a_{k_1}^+a_{k_2}^+a_{k_3}+\cdots\right\}|\Psi_0\rangle$$

1対1対応　　$k_1+k_2-k_3=k$



$\Gamma_k^*\sim T^2$ 温度



$a_k^+|\Psi_0\rangle$ は2つの部分からなる

① z_k の coherent part (準粒子)

② back ground incoherent part

1次元 Luttinger liquid $z_k=0$

No. 102

Date 94 . 10 . 26

$$G(k,t) = -i\langle \Psi_0 | T\, a_k(t)\, a_k^\dagger(0) | \Psi_0 \rangle$$

$$= -i\langle \Psi_0 | a_k(t)\, a_k^\dagger(0) | \Psi_0 \rangle \theta(t) + i\langle \Psi_0 | a_k^\dagger(0)\, a_k(t) | \Psi_0 \rangle \theta(-t) \quad \text{これを} t \text{で微分}$$

$$i\frac{\partial}{\partial t} G(k,t) = \langle \Psi_0 | T\left\{ \frac{\partial a_k(t)}{\partial t} a_k^\dagger(0) \right\} | \Psi_0 \rangle + \langle \Psi_0 | \underbrace{a_k(0)\, a_k^\dagger(0) + a_k^\dagger(0)\, a_k}_{1} | \Psi_0 \rangle \delta(t)$$

$$\left(i\frac{\partial}{\partial t} - \varepsilon_k^0 \right) G(k,t) = \delta(t)$$

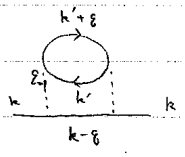
$$a_k(t) = e^{-i\varepsilon_k^0 t}$$

$$G(k,t) = \frac{1}{2\pi}\int d\omega\, G(k,\omega) e^{-i\omega t} \rightarrow \boxed{(\omega - \varepsilon_k^0) G(k,\omega) = 1} \quad \text{free の Green fn.}$$

一般に

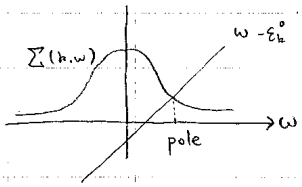
$\Sigma(k,\omega)$ 自己エネルギー

$$\left(\omega - \varepsilon_k^0 - \Sigma(k,\omega) \right) G(k,\omega) = 1$$



$-\mathrm{Im}\,\Sigma(k,\omega) = \Gamma_k$ 減衰

$$G(k,\omega) = \frac{1}{\omega - \varepsilon_k^0 - \Sigma(k,\omega)}$$



pole は $\omega - \varepsilon_k^0 - \Sigma(k,\omega) = 0$ を満たす ω で与えられる.

$$= \frac{z_k}{\omega - E_k^* + i\Gamma_k^* \mathrm{sgn}(k - k_F)} + G_{\mathrm{inc}}(k,\omega)$$

incoherent

$$\Sigma(k,\omega) = \Sigma(k,0) + \left.\frac{\partial \Sigma(k,\omega)}{\partial \omega}\right|_{\omega=0} \omega - i\Gamma_k$$

と近似できるとして

$$\omega\left(1 - \left.\frac{\partial \Sigma}{\partial \omega}\right|_{\omega=0}\right) - \varepsilon_k^0 - \underbrace{\Sigma(k,0)}_{\text{real}} + i\Gamma_k = 0$$

$$\omega = E_k^* - i\Gamma_k^*$$

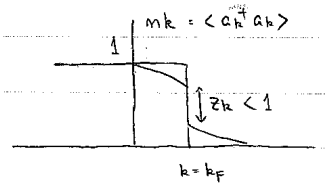
$$E_k^* = \frac{1}{1 - \left.\frac{\partial \Sigma}{\partial \omega}\right|_{\omega=0}} \left(\varepsilon_k^0 + \Sigma(k,0) \right)$$

$$\Gamma_k^* = \frac{1}{1 - \left.\frac{\partial \Sigma}{\partial \omega}\right|_{\omega=0}} \Gamma_k$$

$$\mathrm{Res}\, G(k, \omega = E_k^* - i\Gamma_k^*) = \frac{1}{1 - \left.\frac{\partial \Sigma}{\partial \omega}\right|_{\omega = E_k^* - i\Gamma_k^*}} = z_k$$

z_k

$$\rho^* = \frac{1}{z_k}\rho = \left(1 - \frac{\partial \Sigma}{\partial \omega}\right)\rho$$



$$Q_k^\dagger = \sqrt{z_k}\left(a_k^\dagger + c_3 a_{k_1}^\dagger a_{k_2} a_{k_3} + \cdots \right)$$

$$z_k^{-1} = 1 - \left.\frac{\partial \Sigma}{\partial \omega}\right|_{\omega=0} = 10^{2\sim3} \qquad z_k = 10^{-2\sim-3}$$

T≠0　温度 Green 関数 (Thermal Green function)　松原 Green 関数　松原武夫

相互作用表示 Interaction Repressitation

$\hat{H} = \hat{H}_0 + \hat{H}'$ — (1)　$\hat{H}'$ perturbation　$\beta = \frac{1}{k_B T} = \frac{1}{T}$　$k_B = 1$　$0 < \tau < \beta$

1体　多体

$S(\tau) := e^{-(\hat{H}-\mu\hat{N})\tau} = e^{-(\hat{H}_0-\mu\hat{N})\tau} S(\tau)$ — (2)

Schrödinger 表示の演算子 $\hat{A}$ を用いて

$\rightarrow e^{(\hat{H}-\mu\hat{N})\tau} = S^{-1}(\tau) e^{(\hat{H}_0-\mu\hat{N})\tau}$ — (3)

相互作用表示 $A(\tau)$ を次のように定義する

$\hat{A}(\tau) = e^{(\hat{H}_0-\mu\hat{N})\tau} \hat{A} e^{-(\hat{H}_0-\mu\hat{N})\tau}$ — (4)

$S(\tau)$ を求めるために (2) をτで微分して

$-(\underbrace{\hat{H}-\mu\hat{N}}_{\hat{H}'}) e^{-(\hat{H}-\mu\hat{N})\tau} = -(\hat{H}_0-\mu\hat{N}) e^{-(\hat{H}_0-\mu\hat{N})\tau} S(\tau) + e^{-(\hat{H}_0-\mu\hat{N})\tau} \frac{\partial}{\partial\tau} S(\tau)$

$\frac{\partial S(\tau)}{\partial\tau} = -\hat{H}'(\tau) S(\tau)$ — (5)

$\hat{H}'(\tau) = e^{(\hat{H}_0-\mu\hat{N})\tau} \hat{H}' e^{-(\hat{H}_0-\mu\hat{N})\tau}$

(5) の解は $S(0) = 1$ より

$S(\tau) = \underbrace{T_\tau}_{\tau\text{の小さい順に右から operator を並べる}} \exp\left\{-\int_0^\tau H'(\tau')\,d\tau'\right\}$ — (6)

$1 - \int_0^\tau H'(\tau_1)\,d\tau_1 + \frac{1}{2}\int_0^\tau\int_0^\tau T_\tau H'(\tau_1)H'(\tau_2)\,d\tau_1 d\tau_2 + \cdots$

$\hat{H}_0$ と $\hat{N}$ は交換するから

$\hat{H}_0(\tau) = \hat{H}_0$, $\hat{N}(\tau) = \hat{N}$　$\tau_1 > \tau_2$ として $S(\tau_1, \tau_2)$ を導入する

$S(\tau_1, \tau_2) = T_\tau \exp\left\{-\int_{\tau_2}^{\tau_1} \hat{H}'(\tau')\,d\tau'\right\}$

(6) の $S(\tau)$ は $S(\tau) = S(\tau, 0)$

$\tau_1 > \tau_2 > \tau_3$ の時　$S(\tau_1, \tau_3) = S(\tau_1, \tau_2) S(\tau_2, \tau_3)$

$\tau_1 > \tau_2$ に対して　$S(\tau_1, \tau_2) = S(\tau_1) S^{-1}(\tau_2)$

自由エネルギー Ω は　$e^{-\Omega/T} = T_r\left\{e^{-(\hat{H}-\mu\hat{N})/T}\right\} = T_r\left\{e^{-(\hat{H}_0-\mu\hat{N})/T} S(\beta)\right\}$ — (7)

$\Omega = \Omega_0 - T \ln \langle S(\beta)\rangle$ — (8)

$\Omega_0 = -T \ln T_r\left\{e^{-(\hat{H}_0-\mu\hat{N})/T}\right\}$ — (9)

$\langle S(\beta)\rangle_0 = \frac{T_r\left\{e^{-(\hat{H}_0-\mu\hat{N})/T} S(\beta)\right\}}{T_r\, e^{-(\hat{H}_0-\mu\hat{N})/T}}$ — (10)

↑ H_0 での熱平均

温度 Green 関数　$G_{\alpha\beta}(\vec{r}_1, \tau_1; \vec{r}_2, \tau_2)$ を field op. $\psi_\alpha(\vec{r}_1)$, $\psi_\beta^+(\vec{r}_2)$ を用いて

$$G_{\alpha\beta}(\vec{r}_1, \tau_1; \vec{r}_2, \tau_2) = \begin{cases} -T_r\left[e^{(\Omega+\mu N-H)/T} e^{(\hat{H}-\mu\hat{N})(\tau_1-\tau_2)} \psi_\alpha(\vec{r}_1) e^{-(H-\mu N)(\tau_1-\tau_2)} \psi_\beta^+(\vec{r}_2)\right] & \tau_1 > \tau_2 \\ \pm T_r\left[\quad\text{〃}\quad e^{-(\quad)(\quad)} \psi_\beta^+(\vec{r}_2) e^{+(\quad)(\quad)} \psi_\alpha(\vec{r}_1)\right] & \tau_1 < \tau_2 \end{cases} \quad (11)$$

+ Fermion
− Boson

No. 104

Date 94 · 11 · 2

$\tau_1 - \tau_2 = \tau$ $\beta = 1/T$ とおく $\quad G(\tau<0) = \mp G(\tau+\beta)$ $\quad -\beta<\tau<\beta$

$\because \tau<0$, $G(\tau) = \pm \mathrm{Tr}\left[e^{(\Omega+\mu N-H)/T}\, e^{-(H-\mu N)\tau}\, \psi_\beta^+(r_2)\, e^{(H+\mu N)\tau}\, \psi_\alpha(r_1) \right] = \mp G(\tau+\beta) \quad \tau+\beta>0$

$e^{\Omega/T}\, e^{-(H-\mu N)(\tau+\beta)}$

Heisenberg 表示の op $\psi_\alpha^H(r,\tau)$, $\psi_\beta^H(r,\tau)$ を用いて

$$\left.\begin{aligned} \psi_\alpha^H(r,\tau) &= e^{(H-\mu N)\tau}\, \psi_\alpha(r)\, e^{-(H-\mu N)\tau} \\ \psi_\beta^H(r,\tau) &= e^{(H-\mu N)\tau}\, \psi_\beta^+(r)\, e^{-(H-\mu N)\tau} \end{aligned}\right\} \quad G_{\alpha\beta}(r_1,\tau_1;r_2\tau_2) = -\mathrm{Tr}\left\{ e^{(\Omega+\mu N-H)/T}\, T_\tau\, \psi_\alpha^H(r_1,\tau_1)\, \psi_\beta^{H+}(r_2,\tau_2) \right\}$$

$$= -\langle T_\tau\, \psi_\alpha^H(r_1,\tau_1)\, \psi_\beta^{H+}(r_2\tau_2) \rangle$$

相互作用 表示の $\varphi_\alpha(r_1,\tau_1)$ $\varphi_\beta^+(r_2,\tau_2)$

$$\left.\begin{aligned} \varphi_\alpha(r,\tau_1) &= e^{(H_0-\mu N)\tau}\, \psi_\alpha(\vec{r}_1)\, e^{-(H_0-\mu N)\tau} \\ \varphi_\beta^+(r_2\tau_2) &= e^{(H_0-\mu N)\tau}\, \psi_\beta^+(r_2)\, e^{-(H_0-\mu N)\tau} \end{aligned}\right\} \text{ を用いて}$$

$$G_{\alpha\beta}(r_1\tau_1;r_1\tau_2) = -e^{\Omega/T}\, \mathrm{Tr}\left\{ e^{-(H_0-\mu N)/T}\, S(\beta)\, S^{-1}(\tau_1)\, e^{(H_0-\mu N)\tau_1}\, \psi_\alpha(\vec{r}_1)\, e^{-(H_0-\mu N)\tau_1}\, S(\tau_1)\, S^{-1}(\tau_2)\, e^{(H_0-\mu N)\tau_2}\, \psi_\beta^+(r_2)\, e^{-(H_0-\mu N)\tau_2}\, S(\tau_2) \right\}$$

$$= -e^{\Omega/T}\, \mathrm{Tr}\left\{ e^{-(H_0-\mu N)\beta}\, S(\beta,\tau_1)\, \varphi_\alpha(r_1,\tau_1)\, S(\tau_1,\tau_2)\, \varphi_\beta^+(r_2\tau_2)\, S(\tau_2) \right\} \qquad \tau_1>\tau_2$$

$\tau = \tau_1 - \tau_2 < 0$ に対して

$$G_{\alpha\beta}(r_1\tau_1;r_2\tau_2) = \pm e^{\Omega/T}\, \mathrm{Tr}\left\{ e^{-(\hat{H}_0-\mu\hat{N})/T}\, S(\beta,\tau_2)\, \varphi_\beta^+(r_2,\tau_2)\, S(\tau_2,\tau_1)\, \varphi_\alpha(r_1,\tau_1)\, S(\tau_1) \right\}$$

まとめて書くと

$$G_{\alpha\beta}(r_1\tau_1;r_2\tau_2) = -\frac{\mathrm{Tr}\left\{ e^{-(H_0-\mu N)/T}\, T_\tau\left\{ \varphi_\alpha(r_1,\tau_1)\, \varphi_\beta^+(r_2\tau_2)\, S(\beta) \right\} \right\}}{\mathrm{Tr}\left\{ e^{-(H_0-\mu N)/T}\, S(\beta) \right\}} = e^{-\Omega/T}$$

$$= -\frac{\langle T_\tau\, \varphi_\alpha(r_1,\tau_1)\, \varphi_\beta^+(r_2,\tau_2)\, S \rangle_0}{\langle S \rangle_0} \qquad S = S(\beta) = T_\tau \exp\left\{ -\int_0^\beta H'(\tau)\, d\tau \right\} \qquad \langle\,\rangle_0 = \mathrm{Tr}\left\{ e^{(\Omega_0+\mu N-H_0)/T} \cdots \right\}$$

H_0 の熱平均

$G(\tau)$ の Fourier 変換

$G(\tau)$ は $2\beta = 2/T$ を周期とするから

$$G(\tau) = T\sum_\ell e^{-i\omega_\ell\tau}\, G(\omega_\ell) \qquad \omega_\ell = \ell\pi T \quad \ell \text{ 整数} \qquad \omega_\ell = \begin{cases} (2n+1)\pi T & \text{Fermion} \\ 2n\pi T & \text{Boson} \end{cases}$$

$$G(\omega_\ell) = \frac{1}{2}\int_{-\beta}^{\beta} e^{i\omega_\ell\tau}\, G(\tau)\, d\tau = \frac{1}{2}\left(1 \mp e^{-i\omega_\ell\beta}\right)\int_0^\beta e^{-i\omega_\ell\tau}\, G(\tau)\, d\tau$$

のみが残る

☺ $G(\omega_\ell) = \frac{1}{2}\left\{ \int_0^\beta + \int_{-\beta}^0 \right\}$ $\qquad \int_{-\beta}^0 d\tau\, e^{i\omega_\ell\tau}\, G(\tau) = \mp\int_0^\beta d\tau'\, e^{i\omega_\ell(\tau'-\beta)}\, G(\tau') = \mp e^{-i\omega_\ell\beta}\int_0^\beta e^{i\omega_\ell\tau}\, G(\tau)$

$\mp G(\beta+\tau)$, $\tau' = \tau+\beta$

自由粒子 $H_0 = \sum_k \varepsilon_k\, a_k^+ a_k$

$$G_k^0(\tau) = -\langle T_\tau\, a_k(\tau)\, a_k^+(0) \rangle$$

$$a_k(\tau) = e^{(H_0-\mu N)\tau}\, a_k\, e^{-(H_0-\mu N)\tau} = e^{-(\varepsilon_k-\mu)\tau}\, a_k$$

$$G_k^0(\tau) = -\langle T_\tau\, e^{-(\varepsilon_k-\mu)\tau}\, a_k a_k^+ \rangle$$

$$= -e^{-(\varepsilon_k-\mu)\tau}\langle a_k a_k^+ \rangle \qquad \tau>0$$

$$= +e^{-(\varepsilon_k-\mu)\tau}\langle a_k^+ a_k \rangle \qquad \tau<0$$

$\langle a_k a_k^+ \rangle = 1 - f(\varepsilon_k) = \dfrac{1}{1+e^{-(\varepsilon_k-\mu)/T}}$

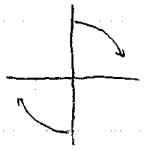
$\dfrac{1}{1+e^{(\varepsilon_k-\mu)/T}}$ ″

$G_k^0(\tau<0) = -G_k^0(\tau+\beta)$

$$\int_0^\beta e^{-i\omega_\ell \tau} G(\tau)\, d\tau = \int_0^\beta e^{-(\xi_k-\mu)\tau} \langle a_k a_k^+ \rangle e^{i\omega_\ell \tau}\, d\tau = \frac{-1}{i\omega_\ell - \xi_k + \mu} \left\{ e^{-(\xi_k-\mu)\beta} \left(e^{i\omega_\ell \beta} \right)^{-1} - 1 \right\} \frac{1}{1+e^{-(\xi_k-\mu)/T}}$$

$\beta\omega_\ell = (2n+1)\pi$

$$= \frac{1}{i\omega_\ell + \mu - \xi_k}$$



No. .....

Date . .